Submission Summary

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Paper Title

Sensor-Based Nadi Pariksha Diagnostic Machine

Abstract

Ayurvedic Pariksha (pulse examination) is a cornerstone of traditional diagnostics, interpreting subtle pulse features (rate, rhythm, strength, temperature) to infer Vata–Pitta–Kapha doshic states. However, manual pulse diagnosis is subjective and skill-dependent. We propose a pre-prototype of a sensor-based Pariksha device that digitizes the Ayurvedic pulse exam. The system uses an array of pressure and photoplethysmographic (PPG) sensors placed at the classical Vata/Pitta/Kapha wrist points (below the thumb) to capture radial pulse waveforms, and a skin temperature sensor to measure Tapamana. Signals are digitized via an embedded microcontroller and analyzed for Ayurvedic parameters (pulse rate, regularity, amplitude, temperature, etc.). Algorithms map these features to dosha dominance (e.g. fast, irregular pulses → Vata; strong, hot pulses → Pitta). A sample UI (smartphone dashboard) displays the pulse waveforms and inferred dosha status. The device will be validated against expert Ayurvedic practitioners (double-blind comparison), aiming for repeatable detection of dosha imbalances. This work aligns with India’s push for integrating AYUSH with digital health (NDHM/ABDM) and preventive care.

Keywords: Ayurveda, Pariksha, Ayurvedic diagnostics, pulse sensor, dosha (Vata/Pitta/Kapha), digital health, AYUSH integration.

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Submission Files

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